

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) Screenshot... 214117.png

Screenshot 2026-07-23 214117.png(image/png) - 288815 bytes, last modified: 7/23/2026 - 100% done
Saving Screenshot 2026-07-23 214117.png to Screenshot 2026-07-23 214117 (2).png

```
import cv2

# Load pre-trained Haar Cascade for face detection
face_cascade = cv2.CascadeClassifier(
    cv2.data.haarcascades + 'haarcascade_frontalface_default.xml'
)

# Open webcam
cap = cv2.VideoCapture(0)

while True:
    ret, frame = cap.read()

    if not ret:
        break

    # Convert frame to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Detect faces
    faces = face_cascade.detectMultiScale(
        gray,
        scaleFactor=1.1,
        minNeighbors=5,
        minSize=(30, 30)
    )
```

```
# Draw rectangle around detected faces
for (x, y, w, h) in faces:
    cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
    cv2.putText(frame, "Person Detected",
                (x, y-10),
                cv2.FONT_HERSHEY_SIMPLEX,
                0.6,
                (0, 255, 0),
                2)

cv2.imshow("Real-Time Surveillance System", frame)

# Press 'q' to exit
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()
```

